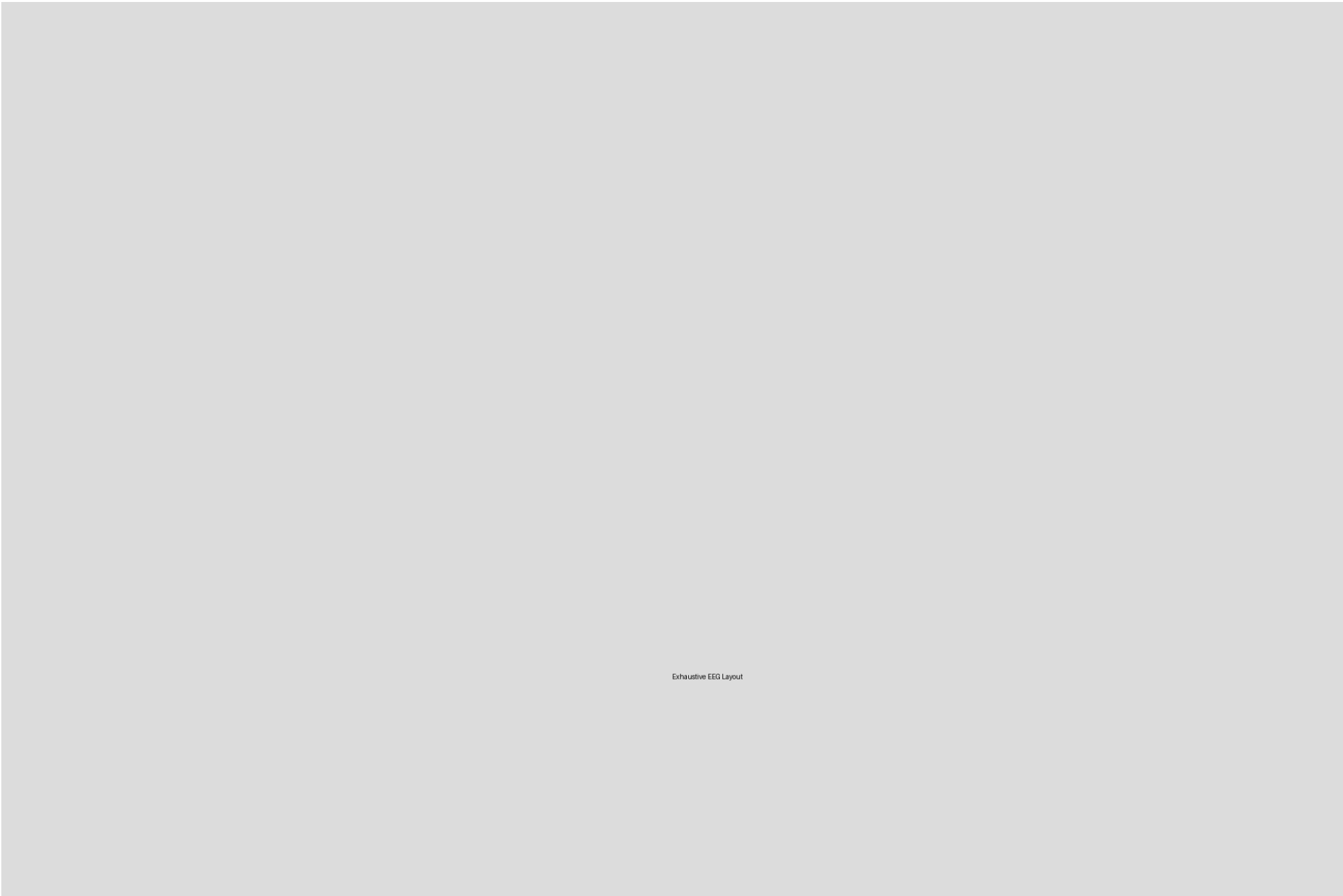


DIY EEG Circuit Guide

EEG Circuit Schematic (Single TL074)

DIY EEG Circuit Guide



Exhaustive EEG Layout

DIY EEG Circuit Guide

DESIGN.md

EEG Circuit Design (Single TL074, Single 9V Battery)

Architecture

1. Virtual Ground (VGND): 4.5V reference using two 10k resistors and a 100uF capacitor.
2. Stage 1: Instrumentation Amplifier: 3-op-amp InAmp, Gain ~ 21 .
3. Stage 2: 50Hz Notch Filter: Passive Twin-T, tuned for 49.8Hz.
4. Stage 3: Gain + Bandpass Filter: Gain ~ 471 , Bandpass 3.4Hz - 34Hz.

Total Gain: $\sim 10,000$.

DIY EEG Circuit Guide

ASSEMBLY.md

Breadboard Assembly Guide

Step 1: Power & VGND

- Red Rail: +9V. Blue Rail: 0V.
- VGND: Junction of two 10k resistors between +9V and 0V. Stabilize with 100uF to 0V.

Step 2: TL074 IC

- Side 1 (Bottom): Pins 1-7. Side 2 (Top): Pins 14-8.
- Pin 4 to +9V. Pin 11 to 0V.

Step 3: Stages

- Stage 1 (InAmp): Pins 12, 13, 14, 5, 6, 7, 2, 3, 1.
- Stage 2 (Notch): Pin 1 to Input, Output to Pin 10.
- Stage 3: Output at Pin 8.

TESTING.md

Safety & Testing

Safety

1. Battery Only.
2. Oscilloscope Isolation.
3. Reference Grounding.

Testing

- Check VGND (4.5V).
- Short inputs to VGND, check output (Flat line).
- Capture Alpha waves (8-13Hz) with eyes closed.

CONNECTIONS.md

Detailed Connection List (Point-to-Point)

Power Supply & Virtual Ground

- Battery (+9V) -> Breadboard Red Rail (+)
- Battery (0V) -> Breadboard Blue Rail (-)
- Resistor Rvg1 (10k) -> Terminal 1: Red Rail (+), Terminal 2: VGND Rail
- Resistor Rvg2 (10k) -> Terminal 1: VGND Rail, Terminal 2: Blue Rail (-)
- Capacitor Cvg (100uF) -> Terminal (+): VGND Rail, Terminal (-): Blue Rail (-)

TL074 IC Connections

- Pin 4 (VCC+) -> Red Rail (+)
- Pin 11 (VCC-) -> Blue Rail (-)
- Pin 10 (Stage 3 Input+) -> VGND Rail (via 1M Rpd resistor)

Stage 1: Instrumentation Amplifier

- Rg (1k) -> Terminal 1: Pin 13, Terminal 2: Pin 6
- R1 (10k) -> Terminal 1: Pin 14, Terminal 2: Pin 13
- R2 (10k) -> Terminal 1: Pin 7, Terminal 2: Pin 6
- R3 (10k) -> Terminal 1: Pin 14, Terminal 2: Pin 2
- R4 (10k) -> Terminal 1: Pin 7, Terminal 2: Pin 3
- R5 (10k) -> Terminal 1: Pin 3, Terminal 2: VGND Rail
- R6 (10k) -> Terminal 1: Pin 2, Terminal 2: Pin 1

Stage 2: 50Hz Notch Filter

- Rn1 (47k) -> Terminal 1: Pin 1, Terminal 2: Node X

DIY EEG Circuit Guide

- Rn2 (47k) -> Terminal 1: Node X, Terminal 2: Node Y (Notch Out)
- Cn1/Cn2 (2x 68n in parallel) -> From Node X to VGND Rail
- Cn3 (68n) -> Terminal 1: Pin 1, Terminal 2: Node Z
- Cn4 (68n) -> Terminal 1: Node Z, Terminal 2: Node Y (Notch Out)
- Rn3/Rn4 (2x 47k in parallel) -> From Node Z to VGND Rail

Stage 3: Gain & Filter

- Cc (100n) -> Terminal 1: Node Y (Notch Out), Terminal 2: Pin 10
- Rpd (1M) -> Terminal 1: Pin 10, Terminal 2: VGND Rail
- Rf (470k) -> Terminal 1: Pin 8, Terminal 2: Pin 9
- Cf (10n) -> Terminal 1: Pin 8, Terminal 2: Pin 9
- Rs (1k) -> Terminal 1: Pin 9, Terminal 2: Node W
- Cs (47uF) -> Terminal (+): Node W, Terminal (-): VGND Rail

External Interfaces

- Electrode A -> Pin 12
- Electrode B -> Pin 5
- Reference Electrode -> VGND Rail
- Oscilloscope Probe -> Pin 8
- Oscilloscope Ground -> Blue Rail (-) (Battery Negative)